

Introduction to Quadratic Relationships and Modeling

Jonathan R. Bacolod

Quadratic Relationship

A **quadratic relationship** is a relationship that can be represented by a **quadratic equation** or **quadratic function** of the form

$$y = ax^2 + bx + c$$

where $a \neq 0$.

Independent and Dependent Variables

The **independent variable** is the quantity you choose or control, while the **dependent variable** changes based on it.

In $y = ax^2 + bx + c$, x is independent and y is dependent.

Steps for Modeling a Situation with a Quadratic Equation

Step 1: Read and Identify

Read the problem carefully to identify the known quantities and the unknown quantity.

Step 2: Define a Variable

Define a variable to represent the unknown quantity.

Step 3: Write an Equation

Write an equation that represents the situation.

Step 4: Simplify

Simplify the equation until it is in standard quadratic form whenever possible.

$$y = ax^2 + bx + c$$

Examples

Example 1

The length of a rectangle is 3 m longer than its width. Its area is 40 m^2 . Represent the situation using a quadratic equation.

Answer: Let x = width. Then length = $x + 3$.

$$x(x + 3) = 40$$

Example 2

A square garden has an area of 121 m^2 . If its side length is represented by x , write the quadratic equation.

Answer:

$$x^2 = 121$$

Example 3

A company's profit is modeled by

$$P = -x^2 + 14x - 24.$$

Explain why this is a quadratic relationship.

Answer: The equation has an x^2 term with $a = -1 \neq 0$, so it fits the standard form $y = ax^2 + bx + c$.

Practice Exercises

Practice 1

The length of a rectangle is 5 m longer than its width. Its area is 84 m^2 . Represent the situation using a quadratic equation.

Practice 2

A square garden has an area of 196 m^2 . If its side length is represented by x , write the quadratic equation that models the situation.

Practice 3

A farmer plans to build a rectangular pen beside a wall using 40 m of fencing for the other three sides. Let x be the width. Write the quadratic expression for the area.

Practice 4

A school's ticket sales are modeled by

$$R = -2n^2 + 120n,$$

where n is the number of tickets sold in hundreds. Explain why this is a quadratic relationship.

Practice 5

The base of a triangle is 3 cm more than its height. If its area is 54 cm^2 , represent the situation using a quadratic equation.

Activities

Activity 1

The product of two consecutive positive integers is 342. Represent the situation using a quadratic equation.

Activity 2

A stone is thrown upward. Its height after t seconds is given by

$$h = -5t^2 + 30t + 2.$$

Identify why this situation is modeled by a quadratic relationship.

Activity 3

The product of two consecutive even integers is 440. Represent the situation using a quadratic equation.

Activity 4

The length of a rectangle is 6 cm greater than its width. If its area is 160 cm^2 , write the quadratic equation representing the situation.

Activity 5

The base of a triangle is 5 cm more than its height. If its area is 75 cm^2 , represent the situation using a quadratic equation.

Thank you!